Can Audio LLMs Understand Music? Evidence from a Comprehensive MIR Benchmark

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***Abstract*—While recent advances in audio-text Large Language Models (LLMs) have expanded the horizons of multimodal music understanding, existing evaluation frameworks remain restricted to narrow task domains, multiple-choice paradigms, or ad-hoc metrics that diverge from established Music Information Retrieval (MIR) standards. To bridge this gap, we introduce CMI-Bench, a comprehensive, instruction-tuned benchmark designed to rigorously evaluate audio-text LLMs across 17 tasks and 29 datasets. CMI-Bench unifies clip-level classification, continuous regression, music captioning, and complex sequential prediction (e.g., melody extraction, structure segmentation, and beat tracking) under a natural language instruction-following formulation. Unlike prior benchmarks, CMI-Bench aligns model outputs with standard MIR evaluation toolkits (e.g., `mir_eval`), enabling direct, metric-level comparisons against specialized supervised baselines. To facilitate efficient benchmarking of proprietary models, we develop a dataset-adaptive sampling protocol that cuts evaluation overhead to 18.3% while strictly preserving full-set model rankings. Extensive evaluation of 23 state-of-the-art open-source and proprietary models reveals that while modern audio LLMs excel at high-level semantic reasoning, they consistently lag behind dedicated MIR baselines on temporal and fine-grained tasks. Furthermore, our diagnostics expose significant prompt sensitivity, training set memorization, and acute cultural, gender, and geographical biases across current systems. CMI-Bench establishes an open, standardized foundation to guide the development of robust, music-aware foundation models.**

I. Introduction

The emergence of large language models (LLMs) has reshaped the landscape of natural language processing by enabling general-purpose models to solve a wide variety of tasks through instruction following. This paradigm, where models are trained not just on pre-text corpora but instruction-response pairs, has unlocked new possibilities in model generalization, few-shot learning, and cross-domain reasoning. Supervised fine-tuning (SFT), also known as instruction finetuning, and reinforcement learning from human feedback have further strengthened LLMs' ability to align with human intent [1].

In the context of music, the instruction-following paradigm holds particular promise. Many music-related tasks are naturally multimodal and domain-specific and often lack large-scale annotated data. Instruction-tuned models can generalize to previously unseen problems such as chord generation under rhythmic constraints or personalized music recommendation based on context. Moreover, LLMs support in-context learning, whereby a model adapts its predictions from task descriptions and a small number of demonstrations provided directly in the input prompt, without updating its parameters. offering a flexible path to interact with world music traditions and diverse user preferences without explicit retraining [2].

Recently, a growing number of audio LLMs [3]–[6], extended LLMs with audio encoders and instruction-following capabilities. However, these models have so far been evaluated on limited tasks, relying on caption similarity metrics on datasets [3], [4], single-choice protocols [7]–[9], or multiple-choice question (MCQ) protocols [10]. Despite these, such evaluations fail to capture the complexity of core music information retrieval (MIR) tasks, limited in real-world performance.

This work extends our preliminary conference paper presented at ISMIR 2025 [11]. in which we first explored the formulation of MIR annotations as instruction-following tasks. The present work fully realises this direction through a broader task formulation, expanded datasets and experiments, and a more comprehensive evaluation framework. Specifically, this work makes three key contributions:

- **Unified MIR Instruction Benchmark:** We introduce **CMI-Bench**, unifying 17 core MIR tasks across 29 datasets into an instruction-following paradigm. By mapping both clip-level and complex sequential MIR annotations (e.g., beat tracking, structure segmentation) into structured prompts, we enable systematic evaluation of dense music understanding in audio LLMs.
- **Standardized & Cost-Aware Pipeline:** We build an open-source evaluation suite that parses free-form LLM outputs directly into standard `mir_eval` metrics. Additionally, we propose a dataset-adaptive sampling strategy that reduces evaluation payload by over 80% while strictly preserving model rank consistency ($\tau \geq 0.90$, $\rho \geq 0.95$).
- **Comprehensive Diagnostics & Bias Analysis:** Evaluating 23 open-source and proprietary models, we identify key performance bottlenecks—such as prompt fragility, temporal alignment gaps, and poor continuous value calibration—while exposing systemic cultural, gender, and chronological biases in current audio foundation models.

The code[1] and testset audio[2] are available.

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[1] https://github.com/nicolaus625/CMI-bench/
[2] https://huggingface.co/datasets/nicolaus625/CMI-bench

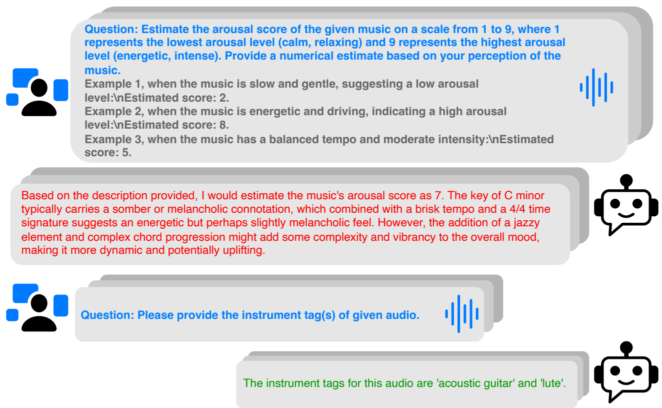


Fig. 1: Instruction-Following Format Data in CMI-Bench and example response from Qwen2-audio.

II. Related Work

A. Instruction Following Datasets

Instruction following refers to the ability of LLMs to perform tasks based on natural language prompts that describe the task itself. This paradigm has become central to recent advances in NLP, where SFT models are trained on a wide range of instruction-response pairs. Super-NaturalInstructions [12] rewrote annotations from over 1,600 diverse NLP tasks into instruction-following formats, showing that models can generalize to unseen tasks given clear instructions. Self-Instruct [13] further advanced this approach by automatically generating diverse instruction–response pairs using the model's own outputs, while Instructional-GPT [14] aligned models with human intent through SFT and reinforcement learning.

These techniques have recently been extended to music, enabling instruction-following models to engage with multimodal and domain-specific tasks. MusicQA [3] and MusicInstruct [4] repurpose descriptions and tags from MIR datasets to generate Q&A pairs. Such a dataset does not distinguish subtasks on instrument, emotion, genre, and caption Q&A pairs, and the evaluating metrics are BERT-score, overestimating models' music understanding capability without fairly comparing with traditional MIR algorithms. Finally, Audio-FLAN [15] presents a large-scale instruction-tuning corpus across more than 80 tasks, unifying understanding and generation in audio, music, and speech. Yet, many tasks are paraphrased to an MCQ format, significantly smaller than the range of pre-defined classes or labels. Furthermore, these works do not provide a model performance benchmark on such tasks, and the evaluation metrics are not compatible with traditional MIR methods.

B. Instruction-Following Benchmarks for Music

While instruction-following has shown great promise in natural language and vision tasks, its application in the music domain remains underexplored. ZIQI-Eval [16] is an instruction following benchmark on textual symbolic music. AIR-Bench [7] covers a broader range of audio types, including music, but emphasizes low-level tasks such as pitch and instrument recognition and relies primarily on MCQ formats. MMAU [10] includes music reasoning, yet only covers six MIR datasets, lacks alignment with MIR-specific evaluation metrics, and reports only average scores across tasks. MuCho Music [8] evaluates music understanding in multimodal models through 1,187 MCQs. AudioBench [17] and MusicBench [18] primarily target audio and text-to-music generation respectively, without addressing MIR tasks. MuChin [19], while valuable for colloquial descriptions, is tailored to Chinese pop song generation. Across these efforts, most benchmarks omit key MIR tasks popularized by MIREX (Music Information Retrieval Evaluation eXchange)[3], an annual community evaluation campaign held in conjunction with the International Society for Music Information Retrieval Conference (ISMIR), , few support sequential tasks, and evaluation protocols often rely on multiple-choice questions rather than the task-specific metrics used in supervised MIR literature.

C. Audio-Textual Large Language Models

Current audio-textual LLMs typically consist of an encoder of speech, audio or music, an intermediate architecture and an LLM backbone. LTU [20] and LTU-AS [21] focus on general audio comprehension and reasoning, combining Whisper speech encoder [22]. MU-LLaMA [3], MusiLingo [4] and Llark [23] are tailored for music-related tasks, leveraging audio encoders and instructional datasets to support captioning and open-ended music question answering. Pengi [24] frames all audio tasks as text generation, unifying audio perception with LLM-based reasoning via a simple prefix-tuning strategy. GAMA [25] and GAMA-IT [25] integrate multi-layer audio features and instruction tuning (CompA-R) to support complex reasoning over general audio, including music. SALMONN-Audio [6] introduces a Q-former window architecture for sequential speech and sound understanding. Qwen-Audio [5] and Qwen2-Audio [26] scale instruction tuning across over 30 audio tasks with hierarchical or natural prompts. Audio-Flamingo [27] and Audio-Flamingo2 [28] incorporate in-context learning and retrieval-based adaptation for audio-text interaction and dialogue. Beyond open-source models, proprietary systems Gemini-3.1 Pro, Gemini-3.5 Flash and GPT- audio may represent the state-of-the-art (SOTA).

III. CMI-Bench

We propose CMI-bench to address the following limitations in evaluating music understanding capabilities of audio-text LLMs mentioned in section II-B. Previous benchmarks often cover only a narrow range of tasks, and no benchmark supports sequential tasks, overlooking many classic tasks which are central to MIR research. Moreover, evaluation protocols are typically inconsistent with standard MIR metrics, making it difficult to compare against traditional supervised models. To address these issues, we reformulate annotations from widely-used MIR datasets into instruction-following prompts and process model outputs into formats compatible with the standard MIR evaluation Python library `mir_eval` [29].

Tasks	Dataset	Metrics	#Test Samples
Key Detection	GS [30]	G-Mean Score	2406
Emotion Regression	EMO [31]	R^2	125
Music Tagging	MagnaTagATune [32]	ROC-AUC, PR-AUC	5329
	MTG-Top50 [33]	ROC-AUC, PR-AUC	11356
Instrument Classification	MTG-Instrument [33]	ROC-AUC, PR-AUC	5115
	NSynth-Instrument [34]	Accuracy	4096
Genre Classification	MTG-Genre [33]	ROC-AUC, PR-AUC	11479
	GTZAN [35]	Accuracy	290
Emotion Tagging	MTG-Emotion [33]	ROC-AUC, PR-AUC	4231
Pitch Estimation	NSynth-Pitch [34]	Accuracy	4096
Singing Technique Classification	VocalSet [36]	Accuracy	1140
Music Captioning	SDD [37]	BLEU, METEOR, ROUGE, BERTScore	1106
	MusicCaps [38]	BLEU, METEOR, ROUGE, BERTScore	2813
Lyrics Transcription	DSing [39]	WER, CER	482
Beat Tracking	GTZAN-Rhythm [35]	F-Measure	290
	Ballroom [40], [41]	F-Measure	685
Downbeat Tracking	GTZAN-Rhythm [35]	F-Measure	290
	Ballroom [40], [41]	F-Measure	685
Melody Extraction	MedleyDB v2 [42]	Melody Accuracy	618
Performance Technique Classification	Guzheng-99 [43]	Micro-F1, Macro-F1	94
Structure Analysis	SongFormBench [44]	Accuracy, HR.5F, HR3F	300
Greek Folk Music Understanding	Lyra-Instrument [45]	Micro-F1, Macro-F1, Exact Match	270
	Lyra-Genre [45]	Micro-F1, Macro-F1, Exact Match	270
	Lyra-Place [45]	Micro-F1, Macro-F1, Exact Match	270
	Lyra-Dance [45]	Accuracy	270
Arab-Andalusian Music Understanding	Arab-Andalusian-Nawba [46]	Accuracy	847
	Arab-Andalusian-Tab [46]	Accuracy	847
	Arab-Andalusian-Form [46]	Accuracy	847
	Arab-Andalusian-Mizan [46]	Accuracy	847

TABLE I: Overview of tasks, datasets, evaluation metrics, and the number of test samples in CMI-Bench.

A. Overview

The CMI-Benchmark encompasses 17 tasks spanning multi-class, multi-label, regression, captioning, and sequential prediction challenges, evaluated across 29 diverse datasets, shown in Table I.. This benchmark integrates traditional MIR tasks with emerging music-and-language objectives, providing a robust platform to assess computational music intelligence. By standardizing splits and metrics, CMI-bench ensures reproducibility and fair comparisons.

B. Self-Instruction of MIR Annotations

In this subsection, we introduce the self-instruction framework for CMI-Bench designed to unify diverse MIR tasks under a consistent NLP paradigm, outlining the design of instructions and input tailored to tasks such as key estimation, genre classification, emotion regression, instrument tagging, and temporal sequence annotations. Our approach leverages structured prompts with multi-class, regression, and sequence-based outputs, enriched with few-shot examples to guide annotation generation.

For multi-label tasks, we allow flexible outputs without providing pre-defined tags, reflecting real-world complexity. For clip-level multi-class tasks with a manageable number of categories, such as musical key estimation and genre and vocal techniques classification, instructions explicitly list all possible choices. For instance, key estimation requires selecting one of 24 major and minor keys, with few-shot examples like "Bb major" to clarify the format. In cases with larger class sets, such as pitch classification on short excerpts across MIDI numbers 9 to 119, we provide a definition of MIDI standard alongside examples (e.g., "A4: 69", "Middle C (C4) = 60") to anchor the task.

Regression tasks, such as arousal estimation, adopt a numerical scale (1 to 9) with descriptive anchors: 1 for "calm, relaxing" and 9 for "energetic, intense." To better utilize LLMs‘ in-context learning capability, we manually create examples for few-shot learning on tie scores to musical characteristics (e.g., "slow and gentle: 2," "energetic and driving: 8"), enabling precise emotional annotation.

Temporal tasks, such as beat tracking and instrument performance technique detection, require structured outputs. Beat tracking outputs timestamps in a comma-separated format (e.g., "0.1s, 1.19s, 2.25s"), while Guzheng (traditional Chinese Kyoto) technique detection uses a Python-style list of tuples (e.g., "[(‘70.8086‘, ‘71.4817‘, ‘Tremolo‘)]"), covering techniques like Vibrato and Glissando. Default outputs "[(‘0.0‘, ‘10.0‘, ‘No Tech‘)]" handle cases with no detections. Melody extraction follows similar principles, balancing specificity and clarity.

[3]https://music-ir.org/mirex/wiki/MIREX_HOME

We forbid tuples to have time overlapping on melody and (down)beat tracking, but allow for playing technique detections.

Inputs are uniformly represented as audio placeholders (e.g., "<|SOA|><AUDIO><|EOA|>"), paired with metadata such as audio paths and time segments. This ensures compatibility with NLP models while preserving MIR task diversity, offering a scalable framework for future efforts.

C. Test Set Sampling Strategy

As shown in Table II, the full CMI-Bench test set contains 61,494 examples across a diverse set of MIR tasks. While full-set evaluation is feasible for open-source models, evaluating proprietary API-based models on the entire benchmark is costly, since inference cost scales with both the number and duration of audio clips. We therefore construct a cost-aware sampled test set for proprietary model evaluation.

The goal of the sampling strategy is to preserve the relative model ranking obtained from the full test set while substantially reducing the number of test examples. For each dataset, we first evaluate open-source models on the full test set and use the resulting model ranking as the reference ranking. We then repeatedly sample random subsets with different sizes and recompute the model ranking using only the sampled examples. The agreement between the subset-based ranking and the full-test ranking is measured using Kendall‘s τ and Spearman‘s rank correlation. Kendall‘s τ measures pairwise ordering consistency between models, while Spearman‘s correlation measures global rank-order agreement.

For each dataset, we select a subset size that reaches high agreement with the full-test ranking, using Kendall‘s $\tau \geq 0.90$ and Spearman‘s $\rho \geq 0.95$ as reliability thresholds. To avoid unstable estimates from extremely small subsets, we additionally enforce a minimum sample size of 100 examples when possible. For datasets with fewer examples than the selected sample size, the full test set is used. This results in a sampled evaluation set of 11289 examples for proprietary models.

Figures 2a and 2b show that most datasets reach stable model rankings with substantially fewer examples than their full test sets. For example, MTT requires only 300 samples out of 5,329 examples, MTG-Top50 requires 500 samples out of 11,356 examples, and MTG-Genre requires 750 samples out of 11,479 examples. More heterogeneous tasks, such as MTG-Emotion and NSynth-Pitch, require larger subsets to preserve ranking stability. Overall, this sampling protocol preserves the evaluation size to approximately 18.3% while maintaining reliable conclusions about relative model performance.

IV. Experiments

A. Evaluation Protocol

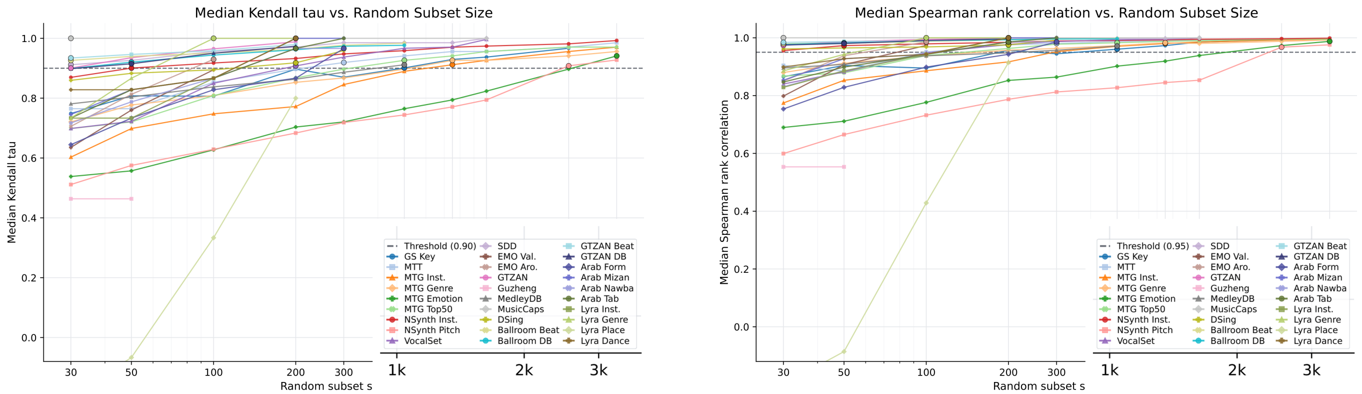
To enable rigorous and fair comparison with traditional MIR systems, we design an evaluation pipeline that closely follows the original task definitions and metrics. All model outputs are automatically post-processed to conform to each task‘s expected format, ensuring compatibility with MIR evaluation tools such as `mir_eval`. Below, we detail the evaluation strategies used for each task category.

TABLE II: Dataset and task sampling statistics.

Dataset / Task	**Full Size**	**Sample Size**	**Preserve**
GiantSteps Key	2406	500	20.8%
MTT	5329	300	5.6%
MTG-Instrument	5115	750	14.7%
MTG-Genre	11479	750	6.5%
MTG-Emotion	4231	2500	59.1%
MTG-Top50	11356	500	4.4%
NSynth-Instrument	4096	100	2.4%
NSynth-Pitch	4096	2000	48.8%
VocalSet-Technique	1140	200	17.5%
SDD	1106	100	9.0%
EMO-Valence	125	125	100.0%
EMO-Arousal	125	100	80.0%
GTZAN	290	100	34.5%
Guzheng-Technique	94	94	100.0%
MedleyDB	618	500	80.9%
MusicCaps	2813	100	3.6%
DSing	482	200	41.5%
Ballroom-Beat	685	100	14.6%
Ballroom-Downbeat	685	100	14.6%
GTZAN-Beat	290	100	34.5%
GTZAN-Downbeat	290	100	34.5%
SongFormBench	300	300	100.0%
ArabAndalusian-Form	847	300	35.4%
ArabAndalusian-Mizan	847	200	23.6%
ArabAndalusian-Nawba	847	200	23.6%
ArabAndalusian-Tab	847	200	23.6%
Lyra-Instrument	270	200	74.1%
Lyra-Genre	270	100	37.0%
Lyra-Place	270	270	100.0%
Lyra-Dance	270	200	74.1%
Total	61619	11289	18.3%

TABLE III: Dataset-adaptive sample sizes for proprietary model evaluation. The sample size is selected according to rank-correlation analysis between subset-based model rankings and full-test model rankings, with a minimum sample size of 100 applied when possible. Preserve denotes the percentage of the full test set retained for evaluation.

1) Classification Tasks: **Multi-Class Classification**. Tasks include short-clip monophonic pitch estimation, instrument classification, singing technique classification, genre classification, Lyra (a type of Greek folk music) dance recognition, and Arab-Andalusian modal/form classification. For the original multi-class tasks, we evaluate using string matching under a case-, space-, and punctuation-insensitive setting: a model‘s response is considered correct if it contains only the correct label and no others. For pitch classification, we additionally require the model to follow the instruction format and return MIDI numbers. For Lyra dance recognition, we treat the task as a binary decision and count a prediction as correct only when the response unambiguously matches the ground-truth yes/no label. For the Arab-Andalusian tasks, models are evaluated on nawba, tab, form, and mizan recognition using accuracy; responses are normalized by removing case, spacing, punctuation, and common diacritic differences before matching the target label.



(a) Kendall's τ (b) Spearman's ρ

Fig. 2: Rank-correlation analysis for dataset-adaptive test set sampling. For each dataset, random subsets of different sizes are sampled and the resulting model ranking is compared with the ranking obtained from the full test set. The dashed lines indicate the reliability thresholds used in our sampling strategy: Kendall's $\tau \geq 0.90$ and Spearman's $\rho \geq 0.95$.

Accuracy is used as the metric for all multi-class classification tasks.

Multi-Label Classification. Tasks include music tagging, genre labelling, emotion tagging, instrument recognition, and the newly added Lyra genre, instrument, and place recognition tasks. As model responses may include synonyms or free-form text, we embed both the predicted and ground-truth tag sets using the GTE-Qwen2 encoder [47], a model optimized for retrieval and multi-label matching. Cosine similarity scores are then used to compute ROC-AUC and PR-AUC for the large-vocabulary music tagging tasks, providing a soft but semantically aligned evaluation. For Lyra, whose annotations are provided as explicit label sets, we instead parse the predicted labels from the normalized response and compute micro-F1, macro-F1, and exact match against the ground-truth label set.

2) Clip-level MIR Tasks: **Key Detection**. We adopt the standard weighted score metric from `mir_eval.key`, which accounts for musically reasonable errors, such as relative minor or parallel key.

Regression. Model outputs are constrained to integers in the range [1, 9]; if a float is returned, we take the floor. Outputs are then z-score normalized to zero mean and unit variance. If a model fails to return a value, we assign the model's mean value. The coefficient of determination (R^2) is computed between predictions and annotations on arousal and valence.

Music Captioning. We assess caption quality using four standard NLP metrics: BLEU [48], [49], METEOR [50], ROUGE [51], and BERTScore [52].

3) Sequential MIR Tasks: **Lyrics Transcription**. We extract lyrics from model outputs by removing typical prefixes (e.g., "lyrics is as follows:"). Word Error Rate (WER) and Character Error Rate (CER) are computed against ground-truth lyrics.

(Down)Beat Tracking. Models are expected to return a list of time points for (down)beat events. We filter non-numeric outputs, sort the list by time, and apply the F-measure metric from `mir_eval.beat`, with a 20ms tolerance.

Melody Extraction. Melody extraction is treated as a sequential regression task on the fundamental frequency of notes calculated by `mir_eval.melody.evaluate` with 50 music cents tolerance. Models are instructed to return a list of (time, pitch) tuples. We discard invalid tuples, such as entries with missing pitches or improperly formatted timestamps. If multiple pitches are predicted for the same timestamp, we use only the first. Evaluation is based on frame-level accuracy.

Song Structure Segmentation. For SongFormBench task, models are instructed to output a sequence of structural segments, each containing a start time, end time, and section label. We parse common list, tuple, dictionary, and free-text interval formats, normalize section names such as verse, chorus, bridge, intro, outro, instrumental, and silence, and discard invalid or non-monotonic intervals. Predicted and reference segmentations are evaluated with `mir_eval.segment`. We report boundary detection F1 at 0.5s and 3.0s tolerances, and additionally compute pairwise F1, Rand Index, Adjusted Rand Index, and boundary deviation for diagnostic analysis.

Instrument Playing Technique Detection. For the GuZheng_99 dataset, we evaluate frame-level predictions using macro- and micro-F1 scores, allowing for overlapping techniques. Invalid predictions, such as incorrect tuple formats, are filtered out. Empty responses are interpreted as a "no technique" prediction covering the full time range.

B. Models

To ensure broad model coverage, we evaluate 23 systems: 17 audio–text LLMs and three omni-modal LLMs with publicly available weights, together with three proprietary models, as summarised in Table IV. The suite spans 323M to 30B parameters and includes general-audio, music-specialised, speech-aware, instruction-tuned, and reasoning-oriented architectures. This diversity enables a controlled comparison of how model scale, modality coverage, and music-specific training affect instruction following across MIR tasks.

Model	#Params	Sound	Music	Speech
Pengi [24]	323M	✓	✓	✗
Audio Flamingo [27]	2.2B	✓	✓	✗
LTU [20]	7B	✓	✓	✗
LTU-AS [21]	7B	✓	✓	✓
MusiLingo-long [4]	7B	✗	✓	✗
MuLLaMA [3]	7B	✗	✓	✗
GAMA [25]	7B	✓	✓	✗
GAMA-IT [25]	7B	✓	✓	✗
SALMONN [6]	13B	✓	✓	✓
Qwen-Audio-Chat [5]	8.4B	✓	✗	✓
Qwen2-Audio-Instruct [26]	8.4B	✓	✓	✓
Audio Flamingo 2 [28]	7B	✓	✓	✓
Audio Flamingo 3 [53]	7B	✓	✓	✓
Audio-Reasoner [54]	7B	✓	✓	✓
Music Flamingo [55]	7B	✗	✓	✗
AudSemThinker [56]	7B	✓	✓	✓
MOSS-Audio [57]	8B	✓	✓	✓
Baichuan-Omni [58]	11B	✓	✓	✓
Qwen2.5-Omni [59]	7B	✓	✓	✓
Qwen3-Omni [60]	30B-A3B	✓	✓	✓
GPT Audio	Undisclosed	✓	✓	✓
Gemini 3.1 Pro	Undisclosed	✓	✓	✓
Gemini 3.5 Flash	Undisclosed	✓	✓	✓

TABLE IV: Comparison of audio-language models by parameter count and training-domain coverage. ✓ denotes reported coverage of a domain, while ✗ denotes that the domain is not explicitly covered. Parameter counts for proprietary models are not publicly disclosed.

V. Results and Discussion

A. Benchmarking Results

Experiment results reveal several important observations about the current state of audio-text LLMs on MIR tasks.

1) LLMs Underperform Traditional MIR Baselines.: Despite LLMs having achieved excellent results on music captioning and multi-choice QA, [4], [8]–[10], [38], , most evaluated models remain substantially behind task-specific supervised systems on standard MIR metrics. The largest gaps occur in fine-grained classification and temporally structured tasks, including note-pitch classification, melody extraction, and beat and downbeat tracking. This suggests that general-purpose instruction following does not yet provide the precision or task-specific inductive biases acquired by dedicated MIR models. Nevertheless, this trend is not universal. Qwen3-Omni exceeds the cited reference result on GTZAN genre classification, while several models match or surpass earlier results on individual music-captioning metrics. Thus, current audio–LLMs can perform competitively on selected semantic tasks, but these strengths do not transfer consistently across the broader MIR task space.

2) Best Performance May Skew toward Training Set: Interestingly, the peak performance on each task is typically achieved by models whose datasets overlap significantly with their training corpora, revealing limited generalization, especially early models in appendix-best column. Qwen2-Audio performs best on MTG-Jamendo-related tasks such as MTG-top50, MTG-Emotion, and SDD captioning, while common on other tagging and caption datasets. This aligns with its use of MTG-Jamendo and FMA during model development via AIR-Bench, suggesting unsatisfying generalization capability. Besides, MusiLingo performs best on MusicCaps, the same dataset it was trained on for captioning and Q&A. Lastly, GAMA shows the best on MTT and NSynth-instrument and comparative on MusicCaps, while common on other datasets on same tasks, reflecting bias in its SFT corpus. These demonstrate that supervised instruction-tuned models can capture task-specific patterns well when training data is directly aligned, but their generalization to unseen or structurally different tasks remains limited.

3) All Models Perform Poorly on DSing Transcription: Despite the absence of instrumental accompaniment and use of English lyrics, none of the models reach usable performance levels on DSing for lyrics transcription except Qwen3-Omni and Audio-Flamingo3, though it is relatively clean. The weak performance of earlier models such as LTU and SALMONN is notable because they incorporate Whisper-based audio encoders. This finding indicates that an ASR-capable encoder alone is insufficient: instruction tuning, output decoding, and exposure to lyric-specific transcription formats are also critical. In contrast, LyricWhiz [76] combines Whisper transcription with GPT-4-based post-processing and reports performance close to specialised systems, suggesting LLM-based agentic explicit correction and formatting stages remain beneficial for MIR.

4) Prompting Format May Impact Performance.: Prompting without task-specific tokens used during training significantly degrades performance. Qwen-Audio performs far worse on Nsynth-Pitch than reported in its original paper. This is likely due to the absence of structured task tokens (e.g., “<|pitch|><|midi_pitch|>piano“) in our prompt. Instead, CMI-bench relies on general natural language instructions. This highlights a critical gap in current audio LLMs: without clearly defined prompting schemas, their ability to interpret instructions can be fragile and fail to generalize. While different prompts for MusiLingo do not provide a significant difference on MusicCaps. Further prompt-format analyses are presented in V-C.

5) Sequential MIR Tasks Remain Challenging.: Tasks requiring temporally structured outputs, including melody extraction, instrument-performance-technique detection, and beat and downbeat tracking, remain difficult for nearly all evaluated models. Although Qwen3-Omni and Gemini 3.1 Pro show improvements on selected beat- and structure-related metrics, no model consistently approaches specialised MIR systems across sequential tasks. Two factors may contribute to this limitation. First, sequence annotations can be represented using heterogeneous formats, including timestamps, frame indices, event tuples, and interval boundaries. Small formatting errors may therefore lead to substantial metric degradation. Second, current audio–text LLMs appear to have limited exposure to densely aligned temporal supervision during pre-training and instruction tuning. Training with timestamped annotations and explicitly constrained decoding formats may improve both output validity and temporal precision.

6) Emotion Regression Fails for All Models.: Continuous emotion prediction remains unstable across models. Several systems obtain negative R^2 values for arousal or valence, indi-

		Qw2.5	Qw3	AF3	MF	MOSS	AST	GPT-A	Gem. Pro	Gem. Flash	Appendix. Best	SOTA
GS-K	GES ↑	49.91	**63.56**	20.00	40.06	10.52	22.12	0.70*	24.60*	14.98*	9.50 (MusiL.)	74.3 [61]
EMO	aR2 ↑	0.33	0.43	-0.31	-0.05	0.43	-0.18	0.08*	0.52*	**0.54***	0.00 (Pengi)	0.62 [62]
	vR2 ↑	-0.31	**0.43**	-0.65	0.03	0.10	-0.71	0.06*	0.34	0.31	0.00 (Salm./Pengi)	0.76 [63]
MTT	ROC ↑	67.72	78.72	**82.10**	81.59	77.08	75.78	52.10*	55.45*	54.31*	81.21 (Gama)	92.0 [64]
	PR ↑	17.50	27.75	**34.53**	32.91	25.65	23.70	9.71*	13.32*	11.01*	34.26 (Gama)	41.4 [62]
M-G	ROC ↑	69.36	69.57	69.51	**70.95**	70.10	70.33	53.03*	55.96*	58.64*	66.39 (Qw.)	88.0 [63]
	PR ↑	9.47	**10.76**	10.43	10.39	9.58	9.58	5.03*	5.89*	6.84*	9.23 (Qw2.)	20.5 [63]
M-E	ROC ↑	60.88	60.42	54.89	61.92	60.08	**65.90**	50.41*	49.96*	49.74*	60.89 (Qw2.)	78.6 [65]
	PR ↑	5.12	6.59	5.27	6.84	5.51	**8.45**	3.83*	3.59*	3.47*	7.85 (Qw2.)	16.1 [65]
M-I	ROC ↑	60.96	58.04	56.77	56.91	56.61	**64.69**	57.03*	60.95*	59.90*	58.90 (Qw2.)	78.8 [66]
	PR ↑	12.97	13.38	12.99	12.38	11.76	11.80	12.69*	**14.84***	14.18*	12.41 (Qw2.)	22.0 [63]
M-50	ROC ↑	68.12	**71.72**	67.91	66.68	67.74	65.95	56.22*	70.71*	71.64*	64.64 (Qw2.)	84.3 [65]
	PR ↑	17.15	19.84	17.51	15.76	16.22	15.38	13.75*	19.33*	**19.87***	16.54 (Qw2.)	32.1 [65]
GTZ.	Acc. ↑	78.62	**92.76**	89.66	71.72	75.86	66.21	47.00*	88.00*	78.00*	72.07 (Qw2.)	83.9 [67]
VS-T	Acc. ↑	22.46	19.30	20.70	25.70	27.81	18.95	17.36*	**42.50***	21.00*	15.61 (Salm.)	76.9 [68]
NI	Acc. ↑	18.41	15.23	44.60	**75.17**	5.18	3.66	0.00*	20.00*	17.00*	58.37 (Gama)	78.2 [69]
NP	Acc. ↑	3.27	2.83	18.58	**39.75**	0.15	0.00	0.90*	10.75*	11.25*	5.74 (Pengi)	89.2 [65]
SDD	BL. ↑	**20.24**	11.37	16.94	16.07	7.50	3.87	7.19*	19.09*	15.29*	**23.40 (Qw2.)**	-
	ME. ↑	17.70	19.50	18.95	17.52	14.98	9.58	7.84*	19.41*	**19.79***	**23.21 (Qw2.)**	16.7 [70]
	RO. ↑	23.69	**36.40**	27.87	26.00	36.20	34.68	11.28*	26.03*	29.71*	30.15 (MusiL.)	111.9 [70]
	BS. ↑	86.06	87.22	87.09	86.61	85.93	84.67	67.92*	87.22*	**87.32***	**87.44 (Qw2.)**	86.0 [70]
MC	BL. ↑	4.93	13.69	27.98	**31.56**	21.67	8.49	6.21*	3.19*	8.14*	21.50 (MusiL.)	21.7 [4]
	ME. ↑	9.32	22.05	23.61	**23.84**	13.75	14.80	5.24*	7.28*	8.93*	22.49 (MusiL.)	22.4 [70]
	RO. ↑	10.11	**33.94**	23.36	24.84	14.75	27.88	4.95*	7.66*	9.16*	30.29 (MusiL.)	30.8 [4]
	BS. ↑	84.66	86.35	**87.54**	86.80	85.98	83.83	79.31*	83.59*	84.69*	85.75 (MusiL.)	87.8 [70]
DS	WE. ↓	82.24	**13.85**	21.29	100.34	121.24	5349.59	163.34*	51.10*	334.72*	115.70 (Qw.)	12.99 [71]
	CE. ↓	57.17	**8.52**	15.18	95.26	115.04	6042.07	180.29*	53.87*	297.83*	96.20 (Qw.)	-
G-B	FM. ↑	15.79	25.98	28.73	24.16	27.48	21.93	0.00*	**30.32***	25.93*	23.69 (Qw.)	88.3 [68]
G-D	FM. ↑	5.28	10.44	**11.89**	9.16	11.65	9.09	0.00*	6.51*	7.27*	10.21 (Qw.)	54.1 [72]
BR-B	FM. ↑	17.83	29.08	28.72	25.96	28.87	25.24	0.00*	**30.41***	26.43*	21.96 (Qw.)	96.8 [73]
BR-D	FM. ↑	7.39	12.53	13.26	10.67	**13.50**	10.56	0.00*	7.63*	8.31*	10.68 (Qw.)	94.1 [73]
MDB	Acc. ↑	1.83	3.48	1.80	2.76	**7.18**	1.96	0.00*	1.16*	0.71*	5.06 (Qw2.)	72.3 [74]
GZ	maF1 ↑	0.00	0.00	0.00	0.00	0.00	0.03	0.00	3.44	**3.46**	**3.18 (Qw2.)**	90.0 [75]
	miF1 ↑	0.00	0.00	0.00	0.00	0.00	0.01	0.00	4.18	**4.24**	**0.89 (Qw2.)**	80.4 [75]
SFB	B-F1@0.5 ↑	6.91	9.37	3.52	13.61	23.77	5.92	4.16*	**33.83***	24.98*	–	70.3 [44]
	B-F1@3.0 ↑	28.61	47.35	9.21	26.17	55.10	23.66	26.78*	**69.63***	55.89*	–	78.4 [44]
Lyra	Inst. AUC ↑	56.57	57.37	50.09	53.92	53.71	55.04	55.42*	**68.20***	63.27*	–	84.6 [45]
	Genre maF1 ↑	0.51	1.55	0.06	5.12	10.04	11.50	17.74*	**38.09***	34.81*	–	39.9 [45]
	Genre miF1 ↑	6.41	7.93	0.71	46.86	15.44	17.22	39.86*	**58.71***	52.33*	–	87.2 [45]
	Place maF1 ↑	0.00	3.81	0.00	5.31	4.86	4.73	11.48*	**23.23***	17.65*	–	34.4 [45]
	Place miF1 ↑	0.00	4.04	0.00	3.14	5.12	6.97	11.00*	**29.69***	23.03*	–	42.4 [45]
Arab	Nawba Acc. ↑	6.52	5.76	11.73	7.13	18.44	**27.66**	8.00*	17.12*	18.02*	–	–
	Tab Acc. ↑	13.04	5.76	17.49	6.04	19.04	**25.65**	4.95*	18.92*	15.32*	–	–
	Form Acc. ↑	20.89	7.82	5.14	3.02	9.62	9.82	9.04*	23.26*	**38.37***	–	–
	Mizan Acc. ↑	20.53	9.88	7.00	5.31	13.63	27.86	9.62*	45.95*	**48.65***	–	–

TABLE V: Performance of the audio-text LLM suite on CMI-Bench datasets. Models include Qwen2.5-Omni (Qw2.5), Qwen3-Omni (Qw3), Audio-Flamingo 3 (AF3), Music-Flamingo (MF), MOSS-Audio-8B (MOSS), AudSemThinker (AST), GPT Audio (GPT-A), Gemini 3.1 Pro (Gem. Pro), and Gemini 3.5 Flash (Gem. Flash). Best completed scores among the evaluated models are shown in bold; * marks proprietary-model scores computed on sampled subsets; the Appendix Best column gives the best measured score from earlier models in Appendix and ISMIR conference paper, suite with corresponding model in parentheses.

cating performance below a mean-value predictor. Their outputs often concentrate near the centre of the rating scale rather than reflecting the variation in the reference annotations. Qwen3-Omni and the Gemini models obtain meaningful positive results, with Gemini 3.5 Flash achieving the highest arousal R^2 and Qwen3-Omni the highest valence R^2 among the evaluated models. However, both remain below the corresponding task-specific results. These findings suggest that current audio–text LLMs can capture coarse emotional tendencies but remain unreliable for calibrated continuous prediction.

7) Generalization to World-Music Tasks Remains Limited.: The results on GZ, Lyra, and Arab further expose limitations in culturally diverse and under-represented music domains. On GZ, nearly all open-source models produce macro- and micro-F1 scores close to zero, while the strongest proprietary models remain far below the cited task-specific results. This indicates that general acoustic and language pre-training does not automatically provide the fine-grained cultural or performance knowledge required by the benchmark. Gemini 3.1 Pro performs most strongly across the Lyra tasks and approaches the cited result for genre macro-F1. However, substantial gaps remain in instrument recognition, genre micro-F1, and place-related prediction. Results on Arab are similarly uneven: AudSemThinker performs best on Nawba and Tab classification, whereas Gemini 3.5 Flash performs best on Form and Mizan classification. No single model dominates across these culturally specific taxonomies. These results suggest that world-music understanding requires more than scaling general-purpose audio–text training. Improving performance will therefore require broader geographic and cultural coverage during pre-training, collaboration with domain experts, and evaluation protocols that preserve culturally specific concepts rather than mapping them only onto dominant Western music taxonomies.

Overall, the results demonstrate that current audio–text LLMs are promising for semantic and instruction-aligned tasks but remain substantially less reliable than specialised MIR systems for continuous prediction, fine-grained classification, culturally specific understanding, and structured temporal output. Their performance is strongly influenced by training-data alignment and prompting conventions, highlighting generalization and output control as central challenges for future music foundation models.

B. Culture and Gender Bias

We further analyze the performance of two audio LLM models, Qwen2-Audio and Audio-Flamingo—on, fine-grained instrument, genre, and music tag categories. While both models show competitive results overall, our breakdown highlights notable performance disparities across instrument types, cultural genres, and voice-related tags.

1) Instrument Bias on MTG-Instrument: Both models achieve high scores on piano, reflecting the strong representation of piano in most training datasets. Western instruments such as violin and accordion perform close to the average, suggesting moderate robustness across common musical timbres. However, performance drops significantly on bongo and harmonica — commonly associated with world music. These results point to a persistent bias toward Western instruments and limited generalization to underrepresented timbres in current pre-training corpora.

2) Cultural Genre Imbalance on MTG-Genre: Genre classification results similarly reveal systematic disparities. Both models show relatively strong performance on mainstream Western pop genres (e.g., 80s, 90s), while genres associated with world music (e.g., Bossanova, Celtic, Chanson, Ethno, Latin) and music traditions (e.g., Medieval) consistently fall below average. For example, Audio-Flamingo‘s performance on Bossanova and Chanson drops severely. Qwen2-Audio performs slightly better on some long-tail genres, but still shows considerable degradation. These highlight a lack of cultural and historical diversity in the data used for instruction tuning and model pretraining.

3) Voice Tag Differences on MTT: A detailed comparison on vocal tags reveals an interesting divergence. Audio-Flamingo is consistently better at identifying *female* voices than male voices, indicating a possible gender-related acoustic or annotation bias. In contrast, Qwen2-Audio achieves higher ROC-AUC for *female* tags but lower PR-AUC, suggesting that while the model ranks positive examples correctly, its absolute predictions remain sparse or overconfident. This mismatch implies that Qwen2-Audio is sensitive to class ranking but may lack calibration in estimating tag presence probabilities, an issue worth investigating for fairness and reliability in music model deployment.

C. Ablation Study on Different Prompts and Trials

We conduct an ablation study on prompt design using GAMA and GAMA-IT models across two representative tasks: GTZAN genre classification and EMO arousal regression. Variant Prompts 1 and 2 are evaluated over three runs, and the bars report mean performance with standard variant as error bars. GTZAN results (bottom row) are relatively stable across prompts and have small variance for each prompt in multi-trials, indicating that most genre-related instructions are consistently followed. The low variance suggests robustness to prompt changes. In contrast, EMO-A results (top row) show relative sensitivity to prompt variation, particularly under the GAMA-IT model. This instability stems from a higher rate of invalid or non-responsible generations, which are scored as mean values during evaluation. Consequently, differences in prompt phrasing might lead to large deviations, especially when valid predictions diverge significantly from the mean score.

VI. Conclusion

In this paper, we introduced CMI-Bench, a comprehensive benchmark designed to systematically evaluate the music understanding capabilities of audio-text and omni-modal LLMs across 17 tasks and 29 datasets. By reformulating traditional and emerging MIR annotations into instruction-following prompts and aligning model outputs with standard `mir_eval` metrics, CMI-Bench bridges the long-standing evaluation gap between general-purpose audio-language models and specialized MIR systems. Furthermore, our dataset-adaptive sampling strategy reduces the evaluation payload to 18.3% of the full test set while strictly preserving full-set model rankings, making rigorous benchmarking scalable and cost-effective for proprietary models. Our extensive empirical evaluation of 23 state-of-the-art systems reveals that while modern audio LLMs excel at high-level semantic reasoning and captioning, they consistently lag behind dedicated supervised baselines on fine-grained, temporally structured, and continuous estimation tasks. Self-reflection on our results highlights critical systemic limitations: current architectures display acute sensitivity to prompt formatting, exhibit severe performance degradation on non-Western and under-represented musical traditions, and suffer from implicit biases stemming from corporate pre-training corpora.

Looking forward, closing the performance gap between general audio LLMs and specialized MIR systems requires shifting from naive scaling toward acoustically and culturally informed modeling. Future research should prioritize: (1) integrating densely aligned temporal supervision during pre-training to empower complex sequential prediction (e.g., polyphonic melody extraction and structural segmentation); (2) constructing geographically diverse, expert-annotated corpora to eliminate Western-centric and demographic biases; and

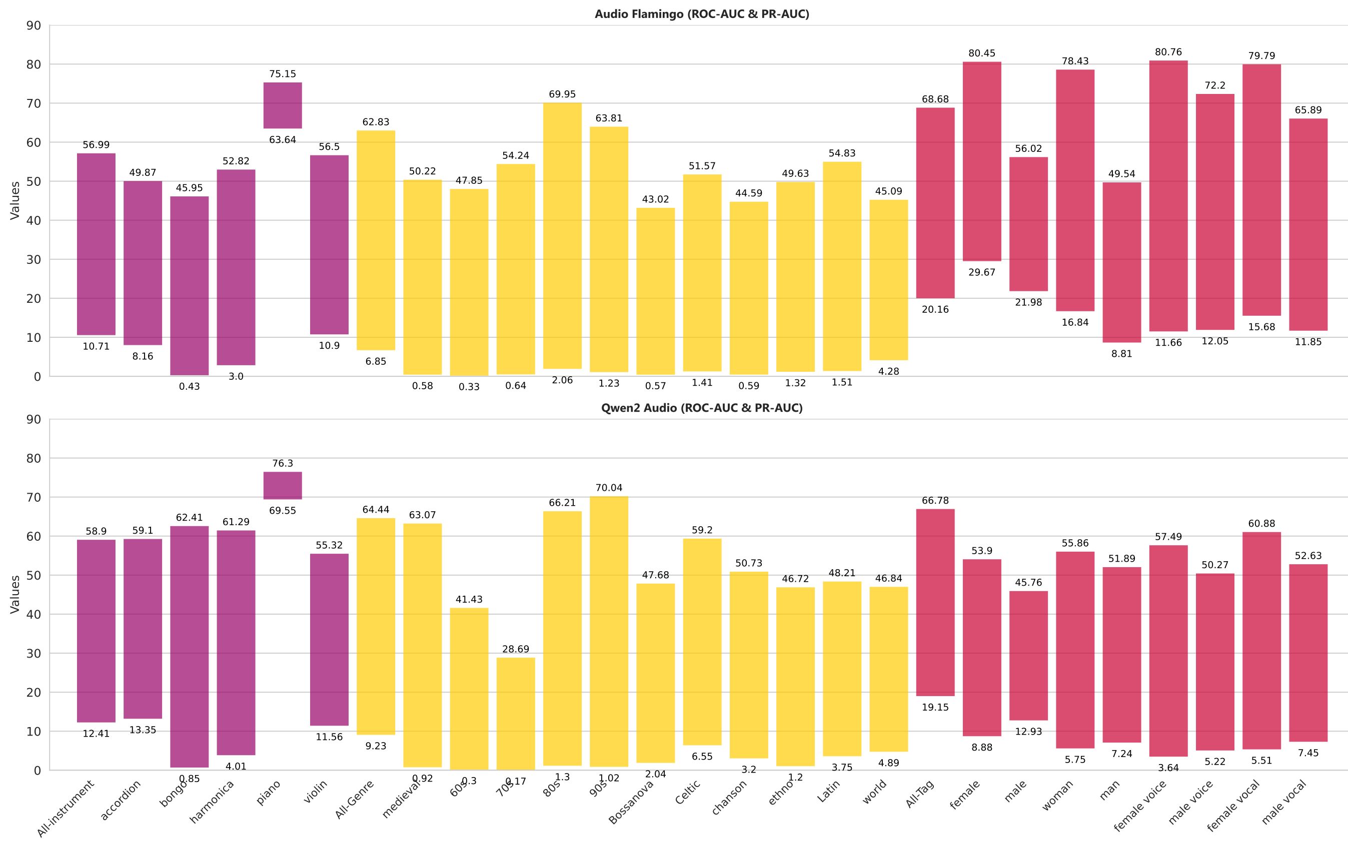


Fig. 3: Fine-grained evaluation of Qwen2-Audio and Audio-Flamingo on instrument (purple), genre (yellow), and vocal (red) tag classification. The upper extremity represents the ROC-AUC value, and the lower is PR-AUC.

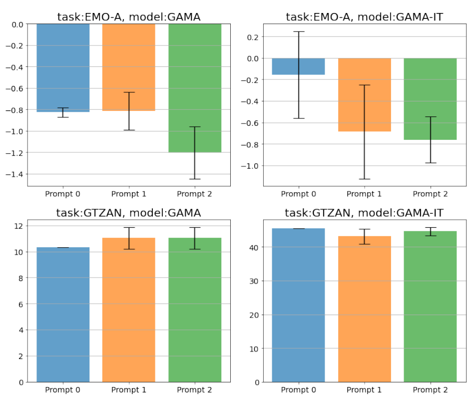


Fig. 4: Ablation Study on Prompt Sensitivity for Genre Classification and Arousal Regression

(3) developing post-training strategies, such as reinforcement learning with MIR-specific reward signals, to enforce strict output formatting and continuous value calibration. We hope CMI-Bench serves as a foundational testbed to guide the next generation of music-understanding foundation models.

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Ethics Statement

CMI-Bench repurposes existing publicly available datasets in the MIR domain by reformatting their annotations into instruction-following formats. No new human annotations were collected, and no human participants were involved in the creation of this benchmark. All data used in the project are licensed under terms that permit non-commercial research use. In compliance with these terms, we license CMI-Bench under a Creative Commons Attribution-NonCommercial-ShareAlike (CC BY-NC-SA) license. To promote long-term accessibility, we host the audio test set on Hugging Face with clear usage restrictions for non-commercial purposes.

The dataset primarily consists of Western, English-language popular music, with limited inclusion of instrumental tracks and non-English songs. Transcription tasks are restricted to English lyrics, and world music instruments besides Guzheng are underrepresented. We acknowledge this cultural and linguistic skew and encourage future extensions to improve global diversity and representation.

This work involves no safety, security, or environmental risks. The benchmark does not require high-compute model training or deployment of potentially harmful generative models. We release CMI-Bench and its evaluation toolkit to foster responsible and reproducible research in audio-language modeling.

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